Mathematics 2, part 4, lecture 1

Gašper Fijavž, gasper.fijavz@fri.uni-lj.si

- Nelder-Mead method
- Linear programming
  - LP toy problem
  - Problems with LP problems
  - LP duality

Nelder-Mead method

- heuristic method performing search in n-dimensional space
- works with samples of $n + 1$ points forming a simplex
- zero-order method
- *can* crawl out of local minima
- a.k.a. *amoeba* method

Simplex?

A *simplex* in n-space is a convex hull of $n+1$ *affinely independent* points.

Points $x_0, x_1, \ldots, x_n$ are affinely independent iff $x_1 - x_0, \ldots, x_n - x_0$ are linearly independent.

Simplex with equal edge lengths?

How to find a simplex with equal edge lengths?

Nelder-Mead method

Notation:

- f ... cost function
- $x_0, x_1, \ldots, x_n$ ... sample points
- $y_0 \leq y_1 \leq \ldots \leq y_n$ ... f values ... $y_i = f(x_i)$
- x_0, x_{n-1}, x_n ... *best, lousy, worst* point

How to produce the next sample of $n+1$ points?

case 1: replace x_n with a better point x_{next} and keep $x_0, \ldots, x_{n-1}$, (REFLECT, EXPAND, INSIDE CONTRACT, OUTSIDE CONTRACT) or

case 2: only keep x_0 and replace all the rest (SHRINK).

Termination criteria

- points $x_0, \ldots, x_n$ *sufficiently* close, or
- f-values $y_0, \ldots, y_n$ *sufficiently* close, or
- use a combination of above.

Nelder Mead examples

Jeklo Ruše example

Jeklo Ruše produces pitchforks and shovels. Manufacturing a single pitchfork takes 3 man-minutes for metalwork and 2 man-minutes woodworking. A shovel is simpler to manufacture, it takes just 1 man-minute for metalwork and again 2 man-minutes of woodworking.

How to optimize production if a pitchfork yields 3 EUR net-income, a shovel yields 2 EUR net-income. We are limited by having only 600 man-minutes for metalwork and 480 man-minutes for woodwork.

Jeklo Ruše example

- Can we do it without pictures?
- Are inequalities really the way to go?

Linear program, standard forms

- What is a linear program?
- What are its properties?
- What is a standard form of a linear program?

Problems with LP

$$\begin{aligned} \max\ & x_1 + x_2 \\ x_1 - x_2 &\leq 5 \\ -x_1 + x_2 &\leq 5 \\ x_1, x_2 &\geq 0 \end{aligned}$$

$$\begin{aligned} \max\ & x_1 + x_2 \\ x_1 + x_2 &\leq -2 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Constructing dual problem

Forget about the pitchfork-shovel optimal solution.

Is there a way to construct an upper bound from first principles?
Is there a way to find a smallest possible upper bound?

Primal and dual problems

For a *primal* LP problem

$$\begin{aligned} \max\ & c^T x \\ Ax &\leq b \\ x &\geq 0 \end{aligned}$$

its *dual* LP problem is

$$\begin{aligned} \min\ & b^T y \\ A^T y &\geq c \\ y &\geq 0 \end{aligned}$$

Sensitivity, meaning of dual variables

- Is there a meaning to dual variables?
- Pulling a rabbit out of the hat.